

Latest Mobile Application Security Threats and Vulnerabilities

Day 01 - Session 01

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- 01** Why Mobile Applications Are Targeted
- 02** Mobile Applications Attack Surface
- 03** Latest Mobile Application Security Vulnerabilities
- 04** Latest Mobile Application Security Threats
- 05** Real-World Attack Scenario
- 06** Protecting Mobile Devices: End-User Best Practices
- 07** Protecting Mobile Applications: Developer Best Practices

Why Mobile Applications Are Targeted ?

1



Handle sensitive
information

2



Compromised Runtime
Environments

3



Are distributed to large
user populations, making
a single weakness highly
impactful

4

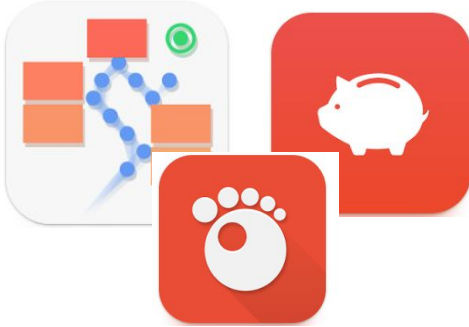


Attackers target both
the application and the
environment in which it
runs

Real-World Threat Scenarios

Goldoson (2023)

A hidden tracker inside 60+ popular apps (100M+ downloads) secretly collected user data and clicked ads without user's consent

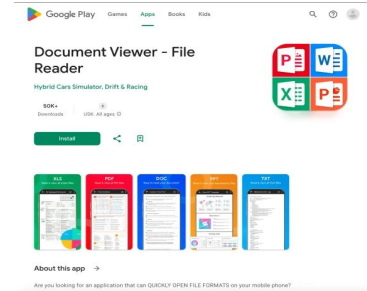


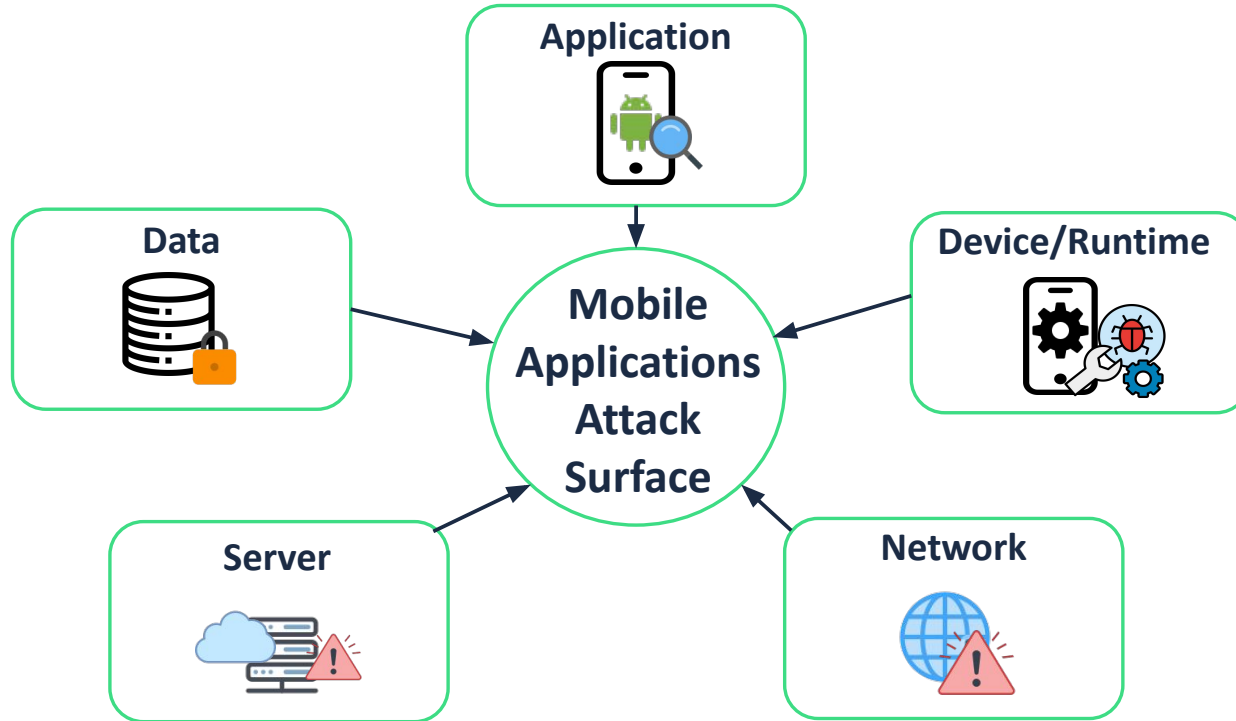
Albiriox (2025)

A malware that lets criminals remotely control a victim's phone and drain bank accounts directly, no stolen password needed

Anatsa (2025)

A fake PDF-reader app stayed harmless for weeks to pass Google's checks, then flipped malicious and stole banking logins from users







Latest Mobile Application Security Threats

01 AI-Assisted Malware & Phishing

02 Insecure APIs and Cloud Misconfigurations

03 Weak Passwords and Session Management

04 Unchecked Open Source Codes & Third-Party SDK

Latest Mobile Application Security Threats

05 Privacy and Compliance Violations

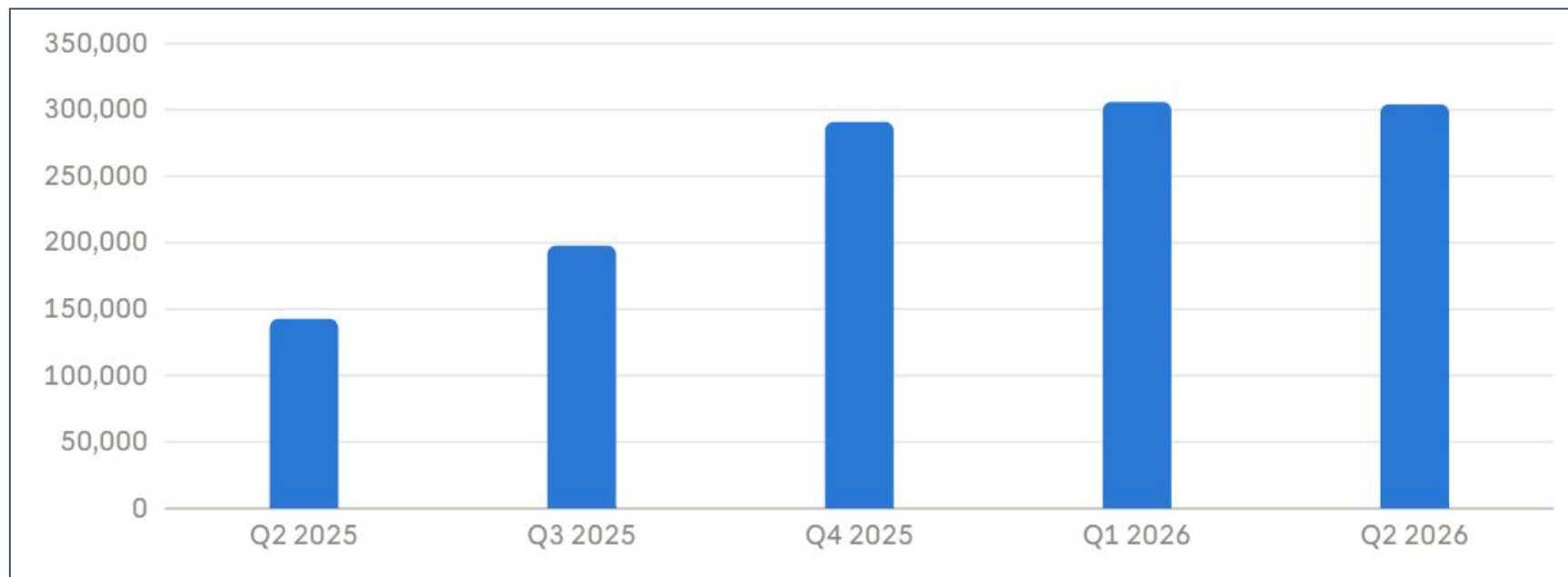
06 Reverse Engineering and Code Tampering

07 Failure in Data Storage & Encryption

08 Broken Access Controls

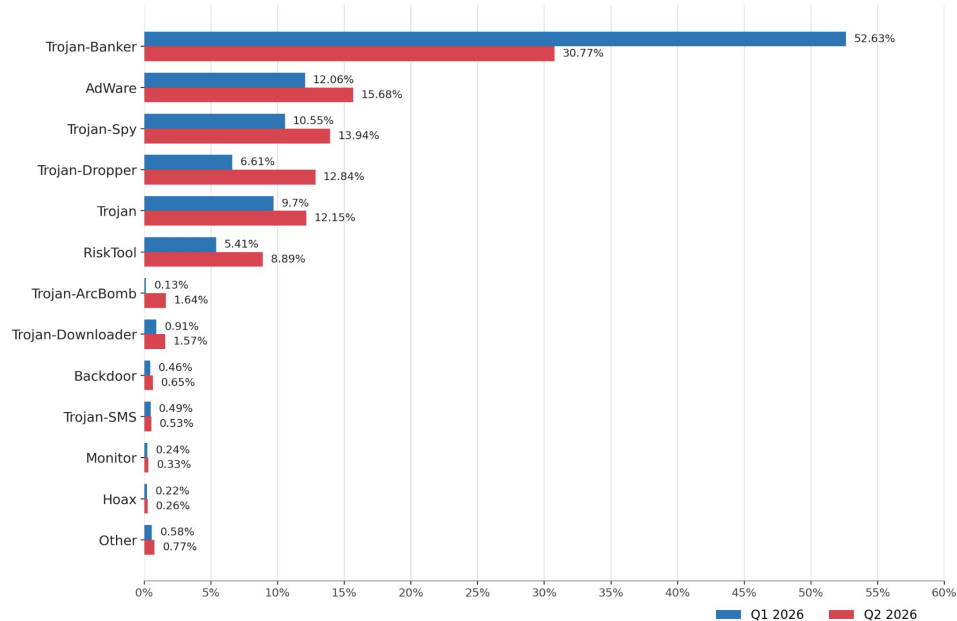
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Detected malicious and potentially unwanted installation packages, Q2 2025 — Q2 2026

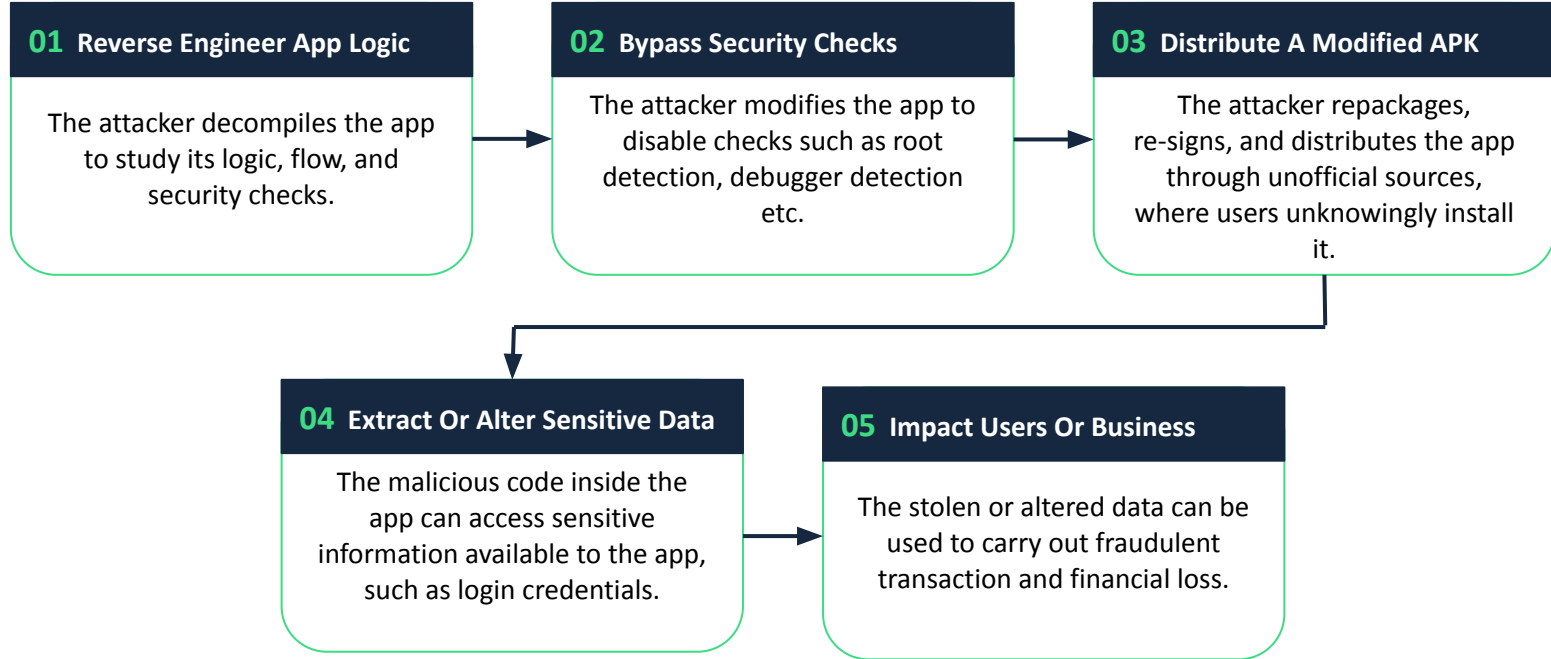


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Threat/Malware Types Detected in Installed Apps (Q1 2026 vs. Q2 2026)



Real-World Attack Scenario



How Can End-Users Protect Their Devices?



How Can Developers Protect Mobile Applications?



- ✓ Use secure coding practices
- ✓ Enforce strong authentication
- ✓ AI-based anomaly detection and anti-phishing safeguards
- ✓ Fix vulnerabilities identified by the security analyst
- ✓ Code obfuscation & RASP
- ✓ Regulatory compliance (GDPR, HIPAA, DPDP Act, BIS)

THANK YOU!!